# Nguyễn Quốc Tuấn

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## Introduction

**AI Engineer** 1+ year of hands-on experience building end-to-end Computer Vision production systems, from model training to real-time deployment on edge devices. Experienced in optimizing and automating slow, manual, or unsafe processes. Experienced on fusing/quantizing model for edge deployment.

## Experience

**Vinsmart Future** Ho Chi Minh City
*AI Engineer* *Aug 2026 – Sep 2026*

***Truck tarp Violation Detection*** | Python, DETR, ResNet

- Built a full violation detection pipeline to flag uncovered construction-material trucks.
- Achieved 91% end-to-end accuracy and 97% recall in automated testing on private datasets.

***Household Re-ID system for Smart-home*** | Python, ViT, PyTorch

- Developed a system assigning IDs to residents, neighbors, and pets to flag unknown persons or animals, finetuning a ViT backbone with a custom retrieval head.
- Achieved 95% mAP and 99% Rank-1 in automated testing on private datasets.

**Rainscales Viet Nam** Ho Chi Minh City
*AI Engineer* *Jun 2025 – Jun 2026*

***Warehouse Inventory Counting System***| Python, ResNet, YOLO, OCR, decord, OpenCV

- AI camera pipeline replacing manual pallet counting, improved worker safety and 5x faster.
- 90–94% video-to-crop accuracy, 81% counting accuracy (measured on onsite production video).

***Event Analytics Pipeline*** | Python, NVIDIA DeepStream, ServiceMaker

- Deployed real-time video analytics on Jetson, handling multiple RTSP streams with a store-and-process pipeline (MP4/JSON chunking, rule engine, rolling buffering), publishing events to Kafka.
- Scaled to 8 cameras at 25fps on Orin Nano and 60 at 25fps (fully loaded) on AGX Orin; also deployed on Jetson Thor for a 5-camera site.

***Model Research & PoC*** | Python, YOLO, ResNet, TensorRT

- Trained a warehouse object detector with 3M images, 16 classes, achieving 85% mAP@0.5.
- Trained a PPE compliance classifier (helmet/vest/coverall), achieving 88% accuracy, false alarm 5%.
- Fused and quantized models into lightweight TensorRT engines for real-time inference on Jetson.

## Education

**Ho Chi Minh City University of Technology – VNUHCMC** Ho Chi Minh City
*Bachelor of Science – Computer Science*

## Research

***SMC2L: Self-Supervised Mixup Contrastive Continual Learning*** | Thesis | Paper

- Proposed a self-supervised continual learning method using contrastive learning for computer vision.
- 92% accuracy on satellite dataset; accepted at IAAA 2026 (publish on Springer LNNS, SCOPUS). Authors: Tin Duc Duong, **Tuan Quoc Nguyen**, Trang Hong Le.

***SAM3-Edge*** | Python, C++, SAM3, TensorRT, NVIDIA DeepStream

- Adapting Meta's SAM3 into DeepStream inference engine, on-the-fly model update.
- Running realtime on 4 camera at 25fps.

## Skills

**Coding:** Python, C/C++, SQL (MySQL, SSMS)
**Machine Learning:** PyTorch, OpenCV, YOLO, ResNet, ViT, OCR
**Edge:** TensorRT, Triton, NVIDIA DeepStream, NVIDIA Jetson (Orin, Thor)
**MLOps:** Kafka, Docker, Linux (Ubuntu)

## Certifications

**IELTS Band 6.5**
**HackerRank SQL (Advanced)**